DIGITAL SAT MATH • UNIT 0.01

Digital SAT Diagnostic & Adaptive Strategy Drill

ANSWER KEY & SOLUTIONS

20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Total Questions: 45 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (45 ITEMS)

Q1 -7	Q2 -6	Q3 -4	Q4 -1	Q5 6
Q6 2	Q7 6	Q8 8	Q9 47	Q10 12
Q11 4	Q12 10	Q13 -4	Q14 0	Q15 6
Q16 43	Q17 2	Q18 21	Q19 2	Q20 4
Q21 B	Q22 B	Q23 D	Q24 D	Q25 C
Q26 B	Q27 C	Q28 A	Q29 B	Q30 B
Q31 A	Q32 C	Q33 A	Q34 B	Q35 D
Q36 B	Q37 C	Q38 A	Q39 B	Q40 C
Q41 D	Q42 C	Q43 A	Q44 D	Q45 C

Part I — Student-Produced Response Diagnostic (Questions 1–20)

1 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $2(x-6)=3x-5$, what is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: -7. Distribute: $2x-12=3x-5$. $-x=-5+12 \Rightarrow x=-7$.

2 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} x+6y=-43 \\ -x-y=13 \end{cases}$$

What is the value of y?

WORKED SOLUTION & VERIFICATION

Answer: -6. Adding the equations eliminates x: $5y=-30 \Rightarrow y=-6$. Then $x=-7$.

3 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 6x-3y=-6 \\ -6x+5y=-2 \end{cases}$$

What is the value of y?

WORKED SOLUTION & VERIFICATION

Answer: -4. Adding the equations eliminates x: $2y=-8 \Rightarrow y=-4$. Then $x=-3$.

4 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $3(x-2) = -3x - 12$, what is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: -1. Distribute: $3x - 6 = -3x - 12$. $6x = -12 + 6 \Rightarrow x = -1$.

5 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $2x + 5 = -5x + 47$, what is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: 6. Applying Linear Equations in One Variable & Systems of Equations principles, $7x = 47 - 5 \Rightarrow x = 6$ Therefore, the verified correct option is 6.

6 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $6(x+1) = -x + 20$, what is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: 2. Distribute: $6x + 6 = -x + 20$. $7x = 20 - 6 \Rightarrow x = 2$.

7 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to $|3x - 9| = 15$?

WORKED SOLUTION & VERIFICATION

Answer: 6. Two cases: $3x - 9 = 15 \implies 3x = 24 \implies x = 8$; and $3x - 9 = -15 \implies 3x = -6 \implies x = -2$. The sum of solutions is $8 + (-2) = 6$.

8 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the positive solution to $|2x - 4| = 12$?

WORKED SOLUTION & VERIFICATION

Answer: 8. $2x - 4 = 12 \implies 2x = 16 \implies x = 8$. (The negative solution is $2x - 4 = -12 \implies x = -4$).

9 SPR Foundational Linear Equations in One Variable & Systems of Equations

A two-digit number has a tens digit t and a units digit u. The sum of the digits is 11. If the digits are reversed, the new number is 27 greater than the original. What was the original number?

WORKED SOLUTION & VERIFICATION

Answer: 47. $t + u = 11$ and $(10u + t) - (10t + u) = 27 \implies 9u - 9t = 27 \implies u - t = 3$. Adding equations gives $2u = 14 \implies u = 7, t = 4$. Original number is 47.

10 SPR Foundational Linear Equations in One Variable & Systems of Equations

A farmer has chickens and cows. In total, there are 30 heads and 84 legs. How many cows does the farmer have?

WORKED SOLUTION & VERIFICATION

Answer: 12. Let c be chickens and w be cows. $c + w = 30 \implies 2c + 2w = 60$. Legs: $2c + 4w = 84$. Subtract: $2w = 24 \implies w = 12$.

11 SPR Foundational Linear Equations in One Variable & Systems of Equations

A store sells notebooks for $3 each and pens for $1.50 each. Maria spent $18 on a total of 8 items. How many notebooks did she buy?

WORKED SOLUTION & VERIFICATION

Answer: 4. Let n = notebooks: $3n + 1.5(8 - n) = 18 \Rightarrow 1.5n = 6 \Rightarrow n = 4$.

12 SPR Foundational Linear Equations in One Variable & Systems of Equations

What is the sum of all solutions to the equation $|2x - 10| = 4$?

WORKED SOLUTION & VERIFICATION

Answer: 10. $2x - 10 = 4$ gives $x = 7$; $2x - 10 = -4$ gives $x = 3$. Sum $= 10$.

13 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 6x - 3y = 0 \\ 5x - 6y = 28 \end{cases}$$

What is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: -4. Solving the system by elimination gives $x = -4$ and $y = -8$.

14 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} 3x - y = -6 \\ -7x - 4y = -24 \end{cases}$$

What is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: 0. Solving the system by elimination gives $x = 0$ and $y = 6$.

15 SPR Foundational Linear Equations in One Variable & Systems of Equations

$$\begin{cases} -4x + 4y = -28 \\ -x - 7y = 1 \end{cases}$$

What is the value of x?

WORKED SOLUTION & VERIFICATION

Answer: 6. Solving the system by elimination gives $x = 6$ and $y = -1$.

16 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $3x - 2y = 9$ and $x + 4y = 17$, what is the value of $5x + 6y$?

WORKED SOLUTION & VERIFICATION

Answer: 43. Multiply the second equation by 2: $2x + 8y = 34$. Add to first: $(3x - 2y) + (2x + 8y) = 9 + 34 \implies 5x + 6y = 43$.

17 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $\begin{cases} ax + by = 12 \\ 4a + 3b = 16 \end{cases}$ and $x = 2, y = 3$, what is the value of a?

WORKED SOLUTION & VERIFICATION

Answer: 2. Substitute $x = 2, y = 3$ into the first equation: $2a + 3b = 12$. Subtract this from $4a + 3b = 16$: $(4a + 3b) - (2a + 3b) = 16 - 12 \implies 2a = 4 \implies a = 2$.

18 SPR Foundational Linear Equations in One Variable & Systems of Equations

If $\begin{cases} 3x - 4y = 7 \\ 2x + 3y = 16 \end{cases}$, what is the value of $x^2 - y^2$?

WORKED SOLUTION & VERIFICATION

Answer: 21. Multiply first equation by 3: $9x - 12y = 21$. Multiply second equation by 4: $8x + 12y = 64$. Add: $17x = 85 \implies x = 5$. Then $2(5) + 3y = 16 \implies 3y = 6 \implies y = 2$. Thus $x^2 - y^2 = 5^2 - 2^2 = 25 - 4 = 21$.

19 SPR Foundational Linear Equations in One Variable & Systems of Equations

In the system $\begin{cases} mx + 3y = 9 \\ 4x + 6y = 18 \end{cases}$, for what value of m does the system have infinitely many solutions?

WORKED SOLUTION & VERIFICATION

Answer: 2. Multiply first equation by 2: $2mx + 6y = 18$. Matching with $4x + 6y = 18$ requires $2m = 4 \implies m = 2$.

20 SPR Foundational Linear Equations in One Variable & Systems of Equations

A barge travels 36 miles upstream in 4 hours and returns 36 miles downstream in 2 hours. What is the speed of the current in miles per hour?

WORKED SOLUTION & VERIFICATION

Answer: 4.5. Let b be barge speed and c be current speed. Upstream: $b - c = 36/4 = 9$. Downstream: $b + c = 36/2 = 18$. Subtract first from second: $2c = 9 \implies c = 4.5$ mph.

Part II — Multiple-Choice Diagnostic Readiness (Questions 21–45)

21 MCQ

If $3x + 14 = 32$, what is the value of x?

- A 46/3
- B 6 ✓
- C 18
- D 7

WORKED SOLUTION & VERIFICATION

Subtract 14 from both sides to get $3x = 18$, then divide by 3: $x = 6$.

22 MCQ

If $2(x - 8) = 8$, what is the value of x?

- A 0
- B 12 ✓
- C 4
- D -4

WORKED SOLUTION & VERIFICATION

Divide both sides by 2: $x - 8 = 4$. Add 8: $x = 12$.

23 MCQ

A moving company charges a one-time fee of $30 plus $55 per mile. If the total charge was $305, how many miles was the move?

- A 6
- B 4
- C 7
- D 5 ✓

WORKED SOLUTION & VERIFICATION

Let n be the number of miles. Then $30 + 55n = 305$, so $55n = 275$ and $n = 5$.

24 MCQ

If $\frac{x}{3} + 5 = 13$, what is the value of x?

- A 27
- B 8
- C 39
- D 24 ✓

WORKED SOLUTION & VERIFICATION

Subtract 5: $\frac{x}{3} = 8$. Multiply by 3: $x = 24$.

25 MCQ

If $8x + 12 = 4x + 36$, what is the value of x?

A 8

B 2

C **6** ✓

D 24

WORKED SOLUTION & VERIFICATION

Subtract $4x$ and 12 from both sides: $4x = 24$, so $x = 6$.

26 MCQ

Suppose that $5x = 15$, what is the value of $10x$?

A 15

B **30** ✓

C 6

D 3

WORKED SOLUTION & VERIFICATION

Multiply both sides of $5x = 15$ by 2: $10x = 30$.

27 MCQ

A taxi company charges a one-time fee of $25 plus $30 per mile. If the total charge was $175, how many miles was the trip?

A 6

B 4

C **5** ✓

D 7

WORKED SOLUTION & VERIFICATION

Let n be the number of miles. Then $25 + 30n = 175$, so $30n = 150$ and $n = 5$.

28 MCQ

Suppose that $\frac{x}{2} + 3 = 9$, what is the value of x?

A **12** ✓

B 6

C 18

D 14

WORKED SOLUTION & VERIFICATION

Subtract 3: $\frac{x}{2} = 6$. Multiply by 2: $x = 12$.

29 MCQ

Suppose that $5x + 14 = 2x + 38$, what is the value of x?

A 24

B **8** ✓

C 10

D 24/7

WORKED SOLUTION & VERIFICATION

Subtract $2x$ and 14 from both sides: $3x = 24$, so $x = 8$.

30 MCQ

Assuming that $4(x-1)=16$, what is the value of x?

- A 4
- **B 5** ✓
- C 3
- D 15

WORKED SOLUTION & VERIFICATION

Divide both sides by 4: $x-1=4$. Add 1: $x=5$.

31 MCQ

A tutoring center charges a one-time fee of $20 plus $60 per session. If the total charge was $260, how many sessions did the student attend?

- **A 4** ✓
- B 3
- C 6
- D 5

WORKED SOLUTION & VERIFICATION

Let n be the number of sessions. Then $20+60n=260$, so $60n=240$ and $n=4$.

32 MCQ

Assuming that $6x+7=3x+37$, what is the value of x?

- A 30
- B 12
- **C 10** ✓
- D 10/3

WORKED SOLUTION & VERIFICATION

Subtract $3x$ and 7 from both sides: $3x=30$, so $x=10$.

33 MCQ

Under the condition that $4x=8$, what is the value of $8x$?

- **A 16** ✓
- B 8
- C 4
- D 2

WORKED SOLUTION & VERIFICATION

Multiply both sides of $4x=8$ by 2: $8x=16$.

34 MCQ

A tutoring center charges a one-time fee of $50 plus $65 per session. If the total charge was $310, how many sessions did the student attend?

- A 6
- **B 4** ✓
- C 5
- D 3

WORKED SOLUTION & VERIFICATION

Let n be the number of sessions. Then $50 + 65n = 310$, so $65n = 260$ and $n = 4$.

35 MCQ

Under the condition that $6x + 8 = 5x + 13$, what is the value of x?

- A 1
- B 5/11
- C 7
- **D 5** ✓

WORKED SOLUTION & VERIFICATION

Subtract $5x$ and 8 from both sides: $1x = 5$, so $x = 5$.

36 MCQ

Provided that $4(x - 9) = 48$, what is the value of x?

- A 39
- **B 21** ✓
- C 12
- D 3

WORKED SOLUTION & VERIFICATION

Divide both sides by 4: $x - 9 = 12$. Add 9: $x = 21$.

37 MCQ

A taxi company charges a one-time fee of $30 plus $40 per mile. If the total charge was $150, how many miles was the trip?

- A 2
- B 5
- **C 3** ✓
- D 4

WORKED SOLUTION & VERIFICATION

Let n be the number of miles. Then $30 + 40n = 150$, so $40n = 120$ and $n = 3$.

38 MCQ

Which values of x satisfy $-2x + 3 > 7$?

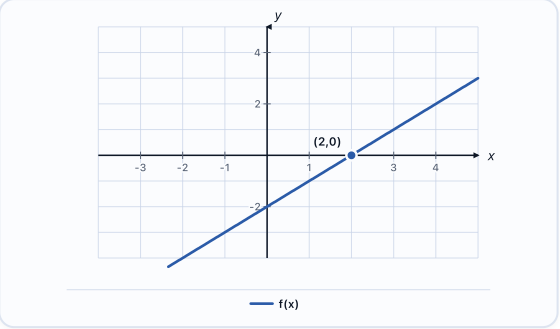
- **A** $x < -2$ ✓
- **B** $x > -2$
- **C** $x > 2$
- **D** $x < 2$

WORKED SOLUTION & VERIFICATION

Applying Linear Functions & Linear Inequalities principles, $-2x > 4 \Rightarrow x < -2$ Therefore, the verified correct option is $x < -2$.

39 MCQ

The graph of a function is shown. At which x-value does the function equal zero?



- **A** $x = 1$
- **B** $x = 2$ ✓
- **C** $x = -2$
- **D** $x = 0$

WORKED SOLUTION & VERIFICATION

The zero (x-intercept) is where the graph crosses the x-axis: $x = 2$.

40 MCQ

A line in the xy-plane has a slope of 3 and passes through the point $(2, 11)$. Which equation represents this line?

- **A** $y = 3x + 11$
- **B** $y = 3x - 5$
- **C** $y = 3x + 5$ ✓
- **D** $y = 2x + 3$

WORKED SOLUTION & VERIFICATION

Using point-slope form: $y - 11 = 3(x - 2) \Rightarrow y = 3x - 6 + 11 = 3x + 5$.

41 MCQ

Which of the following equations represents a horizontal line passing through the point $(3, -5)$?

A $x = 3$

B $y = 3x - 5$

C $y = -5x + 3$

D $y = -5$ ✓

WORKED SOLUTION & VERIFICATION

A horizontal line has slope 0 and an equation of the form $y = c$. Since it passes through $(3, -5)$, the equation is $y = -5$.

42 MCQ

A gym charges a monthly membership fee of $\$30$ plus $\$2$ for each fitness class attended. If a member attends c fitness classes in a month, which expression represents the member's total cost for that month?

A $30c + 2$

B $2(30 + c)$

C $30 + 2c$ ✓

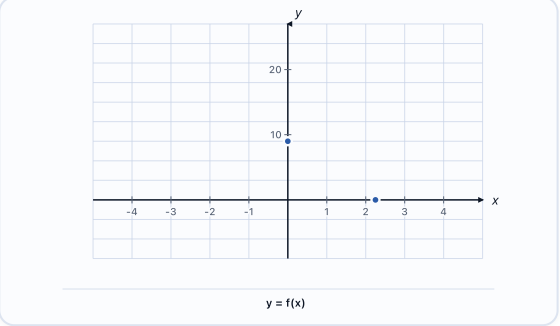
D $\frac{30+c}{2}$

WORKED SOLUTION & VERIFICATION

The base monthly fee is 30 and the variable fee is $2c$, giving $30 + 2c$.

43 MCQ

A line has equation $y = -4x + 9$. What is the y-intercept of the line?



A $(0, 9)$ ✓

B $(-4, 0)$

C $(0, -4)$

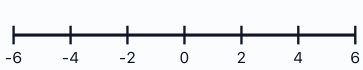
D $(9, 0)$

WORKED SOLUTION & VERIFICATION

In slope-intercept form $y = mx + b$, the y-intercept is $(0, b) = (0, 9)$.

44 MCQ

Which inequality represents all values of x such that $4x - 5 \leq 19$?



A $x \leq 5$

B $x \leq 7$

C $x \leq 4$

D $x \leq 6$ ✓

WORKED SOLUTION & VERIFICATION

Add 5: $4x \leq 24$. Divide by 4: $x \leq 6$.

45 MCQ

What is the slope of the line $2x + 5y = 20$?

A 4

B $\frac{2}{5}$

C $-\frac{2}{5}$ ✓

D $-\frac{5}{2}$

WORKED SOLUTION & VERIFICATION

Solve for y: $5y = -2x + 20 \implies y = -\frac{2}{5}x + 4$. The slope is $-\frac{2}{5}$.